Toward a four-dimensional Yang–Mills construction

Notebook 11: the full flat constraint-current family

Research working document

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Abstract

We bound the nonlinear constraint-current term omitted from the auxiliary family of Notebook 09. The key estimate follows the actual corrected averaging operation through fixed-size blocking steps. Its quadratic contribution at physical spacing h carries a factor h. Flat averages and their physical adjoints are contractions, so these contributions sum geometrically with a bound independent of refinement. This yields both a physical Hilbert form bound and a supremum bound for the constraint vertex. The former gives a uniform complex current neighborhood for the full homogeneous flat source inverse, its coarse precision and its extension. The source coordinate match, first derivative and scope limitations are explicit.

Status. Research proofs and deposited source identifications checked by separate internal derivations and audits. No external certification or novelty claim is made. The result is for $M = L^j$ with a fixed odd $L \geq 3$, which is the refinement sequence used by the blocking construction. The broader odd-M flat Fourier estimates of Notebook 09 remain available as inputs, but are not a claim about an arbitrary nonlinear single-step average of ratio M.

1 Reference operators and the norm

Write D for the fine scalar gradient, d_η for the fine curl, $Q = Q_M^{\mathrm{st}}$ for the normalized straight average, and Q_0 for the scalar block average. Retain the matched source operators

$$G' = (D^*D + Q_0^* a' Q_0)^{-1}, \quad R = I - G' Q_0^* (Q_0 G'^2 Q_0^*)^{-1} Q_0 G', \quad \Lambda = G' R D^*,$$

$$\Delta_0 = d_\eta^* d_\eta + DRD^* + Q^* a Q, \quad G_0 = \Delta_0^{-1}, \quad q_0 = (QG_0Q^*)^{-1}, \quad E_0 = G_0 Q^* q_0.$$

Notebook 09 proves the required inverse, scalar-gauge and auxiliary current estimates in physical cell Hilbert norms, uniformly in refinement and compatible periodic volume. The background is fixed at the identity throughout this notebook. The source value is $a = 1$ at unit final block scale; the distinct scalar penalty a' defines R and is not substituted for a.

For fine or coarse cell blocks use

$$\|T\|_\kappa = \max\left\{ \sup_x \sum_y e^{\kappa d(x,y)} \|T(x,y)\|, \sup_y \sum_x e^{\kappa d(x,y)} \|T(x,y)\| \right\}.$$

Internal Hilbert norms use physical fine volume M^{-d}; coarse cells use counting measure. Distance is coarse-cell ℓ^1 distance or its periodic counterpart. This rectangular operator algebra is submultiplicative and invariant under the fixed bilinear transpose. Choose the same small positive κ as in Notebook 09. The color fiber is fixed; its finite constants are included in those below. Suprema of fine fields mean the supremum of the fixed norm on each positively oriented bond.

2 The separate constraint-current term at flat background

Fix an odd one-step ratio $L \geq 3$ and let $M = L^j$. The fine spacing is $\eta = M^{-1}$ and the final physical block scale is one. We use the corrected nonlinear averaging operation whose flat linearization is Q_M^{st}, the matched scalar source projection R, and the flat inverse and auxiliary-current theorems already proved. The homogeneous source penalty is $a = 1$; the same conclusions hold for a in a fixed compact positive interval. All pairings use physical lattice volume. At complex current, transpose means the fixed bilinear transpose.

The full current inverse in deposited Raw I:13733–13739 is

$$G_1^J = [\Delta_0 - \Theta'_J - \Pi^T \Theta_J \Pi]^{-1}, \qquad \Pi = I - D\Lambda, \tag{1}$$

where $\Lambda = G'RD^*$ is fixed at the flat background. The auxiliary family G_π^J, q_J, E_J uses only Θ'_J and is already controlled. At final scale one $E_J = H^J$, and the separate term is

$$\langle A, \Theta_J B\rangle_\eta = 2\langle C_j^{(2)}(A,B), w_J\rangle_1, \qquad w_J = E_J^T J. \tag{2}$$

Here $C_j^{(2)}$ is the symmetric bilinear quadratic coefficient, not the full second derivative. The argument below controls this actual coefficient by composing fixed L-step jets. It does not replace the nonlinear iterate by a newly chosen single M-step rooted average.

2.1 The corrected coordinate and its one-step jet

The source identifications used here are explicit in the deposit. Raw I:24320–24325 composes the corrected double-bar averaging operation itself. I:24344–24369 gives its logarithmic one-step map and its linear term. I:26985–26988 identifies the tower $C_j^{(2)}$ with the quadratic term of that nonlinear average. The scale ℓ in the source is the retained physical block length, hence $\ell = 1$ here; it is not η.

There is also an explicit input-coordinate bridge. Put

$$\phi(X;c) = \Phi^{\flat}(1, X; c), \qquad \alpha_y(X) = L^{-d} \sum_{x \in B(y)} \Phi^{\mathfrak{g}}(1, X; y, x).$$

The maps on the right are defined using $e^{iX}V$ at I:46965–46973. At $V = 1$, I:47076–47087, explicitly incorporating CMP98 (120), reads

$$e^{i\phi(X;c)} = e^{i\alpha_{c_-}(X)} e^{iF_L(X;c)} e^{-i\alpha_{c_+}(X)}.$$

Thus the corrected unit-angle map on the identity logarithm branch is

$$F_L(X;c) = \frac{1}{i} \log\left[e^{-i\alpha_{c_-}(X)} e^{i\phi(X;c)} e^{i\alpha_{c_+}(X)}\right]. \tag{3}$$

Write subscripts $1, 2$ for homogeneous Taylor coefficients and abbreviate $\alpha_{c_\pm}$ by $\alpha_\pm$. BCH gives

$$\begin{aligned} C_L(X,X)(c) = {} & \phi_2 - \alpha_{-,2} + \alpha_{+,2} \\ & + \frac{i}{2}(-[\alpha_{-,1}, \phi_1] - [\alpha_{-,1}, \alpha_{+,1}] + [\phi_1, \alpha_{+,1}]). \end{aligned} \tag{4}$$

Polarization defines $C_L(A,B)$. In particular, the later unit-lattice vertex named $\Phi_2^{\flat}$ is not substituted for (4); the endpoint contour and BCH terms are retained.

At flat background, the explicit linearization I:24357–24369 has identity transports, $Y = Y_x = 0$ and $g(0) = 1$. Its correction terms vanish, leaving

$$F_L(X) = LQ_L^{\mathrm{st}} X + C_L(X,X) + O(X^3). \tag{5}$$

Every intermediate reference background is again the identity, so the same one-step jet is used at every scale.

For fixed L, the maps in (3) are finite analytic products, means and logarithms near the identity. Consequently there is a finite constant K_L such that

$$|C_L(A,B)(c)| \le K_L \max_{b\in\mathcal{D}_L(c)} |A(b)| \max_{b\in\mathcal{D}_L(c)} |B(b)|, \tag{6}$$

where $\mathcal{D}_L(c)$ is the fixed finite set of input bonds read by the one-step map. The deposited one-step remainder bound I:24349–24353 also supplies such a constant by taking its quadratic coefficient. No bound uniform in $L\to\infty$ is needed.

If the fine physical spacing is h, then the input angle is hA and the output potential is divided by Lh. Therefore

$$\mathcal{F}_h(A) = (Lh)^{-1}F_L(hA) = Q_hA + \mathcal{B}_h(A,A) + O(A^3), \qquad \mathcal{B}_h(A,B) = \frac{h}{L}C_L(A,B). \tag{7}$$

The factor h will make the sum over scales convergent.

2.2 Paired form bounds and the geometric scale sum

Let N_L bound the number of output bonds c whose set $\mathcal{D}_L(c)$ contains any given input bond. This is a fixed geometric constant, with covering-space multiplicities on short tori. Set $k_L = K_LN_LL^{d-1}$.

Lemma 2.1 (The one-step physical form)**.** *For bounded coarse weight w,*

$$|\langle w, \mathcal{B}_h(A,B)\rangle_{Lh}| \le k_Lh\,\|w\|_\infty\,\|A\|_{2,h}\,\|B\|_{2,h}\,. \tag{8}$$

The flat straight average and its physical adjoint satisfy

$$\|Q_hA\|_{2,Lh} \le \|A\|_{2,h}\,, \qquad \|Q_h^*w\|_\infty \le \|w\|_\infty\,. \tag{9}$$

Proof. The coarse pairing contributes $(Lh)^d$, while (7) contributes h/L. Hence its absolute value is at most

$$K_LL^{d-1}h^{d+1}\,\|w\|_\infty \sum_c \Big(\sum_{b\in\mathcal{D}_L(c)}|A(b)|^2\Big)^{1/2}\Big(\sum_{b\in\mathcal{D}_L(c)}|B(b)|^2\Big)^{1/2}.$$

Cauchy–Schwarz over c, the incidence bound N_L, and the fine volume factor h^d prove (8).

For a coarse bond in direction μ,

$$(Q_hA)_\mu(y) = L^{-d-1}\sum_{x\in B(y)}\sum_{r=0}^{L-1} A_\mu(x+rhe_\mu).$$

The coefficients are nonnegative and each row sums to one. For each fine bond there are exactly L string offsets, counting multiplicity, and each fixes one starting point and block center. Its column sum is thus L^{-d}. The physical adjoint multiplies this by the coarse/fine volume ratio L^d. Therefore both Q_h and Q_h^* have nonnegative rows summing to one. Jensen's inequality proves the Hilbert contraction, and the adjoint row identity proves the supremum contraction. This counting also applies on short periodic tori. □

Set $h_t = L^{t-j}$ for $0\le t\le j$. Let Q_t act from spacing h_t to h_{t+1} and put

$$Q_{r:s} = Q_{s-1}\cdots Q_r \quad (r<s), \qquad Q_{r:r} = I.$$

Theorem 2.2 (Uniform quadratic coefficient of the corrected iterate)**.** *The actual quadratic coefficient has the exact expansion*

$$C_j^{(2)}(A,B) = \sum_{t=0}^{j-1} Q_{t+1:j}\mathcal{B}_{h_t}(Q_{0:t}A, Q_{0:t}B). \tag{10}$$

With $k_* = k_L/(L-1)$*,*

$$|\langle w, C_j^{(2)}(A,B)\rangle_1| \le k_* \|w\|_\infty \|A\|_{2,\eta} \|B\|_{2,\eta}, \tag{11}$$

uniformly in j*, refinement and compatible volume.*

Proof. If a partial composition has linear part P and quadratic coefficient C, the next step replaces the latter by $Q_tC + \mathcal{B}_{h_t}(P,P)$. This proves (10) inductively, with no factor two because these are homogeneous coefficients. Pair each summand with w, move the suffix average to w, and apply (8) and (9). The result is bounded by

$$k_L \|w\|_\infty \|A\|_{2,\eta} \|B\|_{2,\eta} \sum_{t=0}^{j-1} h_t, \qquad \sum_{t=0}^{j-1} h_t = \frac{1-L^{-j}}{L-1}.$$

This proves (11); for $j=0$ the coefficient is zero. Computing the finite Taylor jet at zero does not require a common nonlinear neighborhood for the entire iterated map. □

2.3 The vertex as a local Hilbert and supremum operator

Define the symmetric fine operator T_w by

$$\langle A, T_w B\rangle_\eta = 2\langle C_j^{(2)}(A,B), w\rangle_1. \tag{12}$$

The previous theorem gives $\|T_w\|_{2\to 2} \le 2k_* \|w\|_\infty$. The final nonlinear average reads a fixed number of final unit cells. The exact source statement is two final blocks (I:24332–24336); alternatively, a one-step physical radius $\rho_L h$ sums to at most $\rho_L/(L-1)$. Thus $T_w(x,y)$ vanishes when the unit-cell distance exceeds a fixed R_*, independent of j. With $v(R_*) = \#\{z \in \mathbb{Z}^d : |z|_1 \le R_*\}$,

$$\|T_w\|_\kappa \le 2k_* v(R_*) e^{\kappa R_*} \|w\|_\infty =: \tau_\kappa \|w\|_\infty. \tag{13}$$

Here the weighted norm uses physical cell Hilbert block norms. Each block norm is bounded by the global Hilbert norm and only a fixed number of cells is summed. There is no fine-cell dimension factor.

Lemma 2.3 (A stronger bound for the constraint vertex itself)**.** *The operator* T_w *is also uniformly bounded on fine* ℓ^∞*, with fixed coarse-cell range and exponential cell weights. This assertion does not require a supremum bound for* Π*.*

Proof. Write the one-step coefficient in local fine coordinates as

$$C_L(A,B)(c) = \sum_{b,b' \in \mathcal{D}_L(c)} c_{c;b,b'}(A(b), B(b')).$$

Let K_L^{coef} bound these finitely many bilinear coefficient norms and $m_L = \max_c |\mathcal{D}_L(c)|$. The one-step fine operator T_w^h has absolute row sum at most

$$2L^{d-1} h N_L m_L K_L^{\mathrm{coef}} \|w\|_\infty. \tag{14}$$

The factors are the physical volume ratio L^d, the quadratic factor h/L, and the factor two in (12). Each row meets at most N_L outputs and m_L second input bonds per output. Transpose symmetry gives the corresponding absolute column bound.

The exact multilevel operator identity is

$$T_w = \sum_{t=0}^{j-1} Q_{0:t}^* T_{Q_{t+1:j}^* w}^{h_t} Q_{0:t}. \tag{15}$$

Both average factors and the backward weight are supremum contractions. Hence

$$\|T_w\|_{\infty\to\infty} \le \frac{2L^{d-1} N_L m_L K_L^{\mathrm{coef}}}{L-1} \|w\|_\infty . \tag{16}$$

Its fixed final range gives weighted cell-block bounds with supremum fibers at another fixed range cost. The physical weights and all periodic multiplicities were already included in (14). □

2.4 The actual coefficient weight and the full inverse

Use the auxiliary theorem's constants

$$\|G_0\|_\kappa \le g, \qquad \|Q\|_\kappa , \|Q^*\|_\kappa \le b, \qquad \|q_0\|_\kappa \le q, \qquad c = gb^2 q, \quad e = gbq.$$

On its current ball of radius $\varepsilon_{\mathrm{aux}}$, $\|E_J\|_\kappa \le 4e$. A bounded fine current has physical Hilbert norm at most $\sqrt{d}\,\|J\|_\infty$ in each unit cell. The transpose rectangular Schur estimate therefore proves

$$\|w_J\|_\infty = \left\| E_J^T J \right\|_\infty \le 4\sqrt{d}\, e\, \|J\|_\infty . \tag{17}$$

This row sum is absolutely convergent also for globally bounded, non-square-integrable currents in infinite volume. The map $J \mapsto E_J^T J$ is analytic into coarse ℓ^∞, because evaluation is a bounded bilinear map in these norms. Consequently $\Theta_J = T_{w_J}$ is analytic and

$$\|\Theta_J\|_\kappa \le 4\sqrt{d}\, e\tau_\kappa \|J\|_\infty . \tag{18}$$

It has fixed range in its potential arguments, although its coefficient weight depends nonlocally on the current.

Let $p = \|\Pi\|_\kappa < \infty$, using the proved Hilbert bound on $D\Lambda$, and let θ be the auxiliary constant $\|\Theta_J'\|_\kappa \le \theta \|J\|_\infty$. Put

$$\xi = \theta + 4p^2\sqrt{d}\, e\tau_\kappa, \qquad \varepsilon_1 = \min\left\{ \varepsilon_{\mathrm{aux}}, \frac{1}{4g\xi(1+c)} \right\}. \tag{19}$$

The vanishing-perturbation case has the evident interpretation.

Theorem 2.4 (The full homogeneous flat current family)**.** *For $\|J\|_\infty < \varepsilon_1$, the actual full flat operators*

$$G_1^J = [\Delta_0 - \Psi_J]^{-1}, \quad q_1^J = (QG_1^J Q^*)^{-1}, \quad E_1^J = G_1^J Q^* q_1^J, \qquad \Psi_J = \Theta_J' + \Pi^T \Theta_J \Pi \tag{20}$$

are analytic in the weighted physical-cell Hilbert algebra and obey

$$\left\| G_1^J \right\|_\kappa \le 2g, \qquad \left\| q_1^J \right\|_\kappa \le 2q, \qquad \left\| E_1^J \right\|_\kappa \le 4e, \qquad QE_1^J = I. \tag{21}$$

The constants are uniform in $M = L^j$, compatible volume and the specified compact positive penalty interval. Fine derivative sandwiches of G_1^J of total order at most two have the same kind of uniform weighted Hilbert bounds.

Proof. Equations (18) and (19) give $\|\Psi_J\|_\kappa \le \xi \|J\|_\infty$ and $\|G_0\Psi_J\|_\kappa < 1/4$. Thus

$$G_1^J = (I - G_0\Psi_J)^{-1}G_0, \qquad \left\|G_1^J - G_0\right\|_\kappa \le 2g^2\xi \|J\|_\infty .$$

The coarse inverse exists because

$$\left\|q_0 Q(G_1^J - G_0)Q^*\right\|_\kappa \le 2qb^2g^2\xi \|J\|_\infty < \frac{1}{2}.$$

The second Neumann series gives the bounds in (21). Analyticity follows from the two norm-convergent inverse series and the already proved analyticity of Ψ_J.

For a prescribed fine-difference sandwich, the resolvent identity is

$$\begin{aligned}\mathcal{D}_\alpha G_1^J \mathcal{D}_\beta = \mathcal{D}_\alpha G_0 \mathcal{D}_\beta \\ + (\mathcal{D}_\alpha G_0)\Psi_J(I - G_0\Psi_J)^{-1}G_0\mathcal{D}_\beta.\end{aligned}$$

All needed reference sandwiches and one-sided factors are bounded for $\alpha + \beta \le 2$. Derivatives are placed on the outside propagators; none acts on J. □

Corollary 2.5 (Factorial derivatives on smaller current balls)**.** *Let $\mathcal{T}_J$ denote G_1^J, q_1^J or E_1^J, with corresponding uniform bound $M_\mathcal{T} = 2g$, $2q$ or $4e$. For $r = \varepsilon_1 - \|J\|_\infty > 0$ and $n \ge 1$,*

$$\|D^n\mathcal{T}_J[K_1, \dots, K_n]\|_\kappa \le M_\mathcal{T}\, n! \left(\frac{e_{\mathrm{num}}}{r}\right)^n \prod_{i=1}^n \|K_i\|_\infty , \qquad e_{\mathrm{num}} = \exp(1).$$

Here e_{num} is Euler's number, distinct from the extension constant $e = gbq$.

Proof. Apply the multivariable Cauchy estimate to $\mathcal{T}_{J+\sum_i z_i K_i}$ on polydiscs of radii approaching $r/(n\|K_i\|_\infty)$. The coefficient containing every z_i once is the displayed mixed derivative. Its norm is bounded by $M_\mathcal{T} n^n r^{-n} \prod_i \|K_i\|_\infty$. Use $n^n \le e_{\mathrm{num}}^n n!$. Zero directions follow by multilinearity. □

2.5 First response, source geometry, and limitations

The exact first-order constraint form is

$$\langle A, \Theta_K^{[1]} B\rangle_\eta = 2\langle C_j^{(2)}(A, B), E_0^T K\rangle_1. \tag{22}$$

The derivative of E_J is multiplied by the zero base current and does not enter. Since $\Pi E_0 = E_0$ and the auxiliary first coarse-precision derivative vanishes,

$$Dq_1^0[K] = -E_0^T \Theta_K^{[1]} E_0. \tag{23}$$

Equivalently, its bilinear coarse form is $-2\langle C_j^{(2)}(E_0B, E_0C), E_0^T K\rangle_1$. This term is not forced to vanish by the auxiliary cancellation.

The full perturbation Ψ_J is nonlinear in J. Its derivatives therefore do not obey the auxiliary ordered-chain formulas without additional terms. Uniform analytic operator bounds give factorial derivative estimates on smaller current balls. Spatially, already (22) may transport a current from cell x to a coefficient cell y through E_0^T, and then branch along two extension legs to cells b, c. Its natural estimate is a sum of Y-shaped trees over y. An auxiliary ordered path $b \to x \to c$ at the same exponential rate must not be inferred.

There is also a separate pointwise distinction. Lemma 2.3 bounds the constraint vertex itself on fine ℓ^∞. It does not assert that the order-zero projection $\Pi = I - D\Lambda$ is uniformly bounded in that

norm, so it does not by itself prove a supremum bound for $\Pi^T \Theta_J \Pi$ or for the full current inverse. The full-family theorem above uses physical cell Hilbert norms throughout. Notebook 12 separately proves pointwise bounded-forcing estimates on a smaller uniform current ball. It uses the identities $G_0\Lambda^* = DZ$ and $G_0\Pi^* = G_0 - DZD^*$, with $Z = G'RG'$, to keep derivatives off the current and to avoid assuming that Π is bounded on arbitrary supremum-norm input.

A macroscopic path integral need not be bounded uniformly on fine physical L^2: a line of fixed physical length has trace norm of order $\eta^{(1-d)/2}$. The present proof does not invoke such a trace. It confines every path coefficient to the fixed L-step geometry, retains the physical factor h_t, and uses contractions before summing the geometric series. The result therefore does not depend on treating two nonlinear averages with the same linear part as the same map.

The mathematical estimates have separate internal checks. The operator identification uses the exact deposited coordinate, composition and coefficient formulas cited above; the original CMP98 full text has not been independently reverified in this calculation. The result concerns the homogeneous flat source family. It does not establish nonflat current stability, retained multiscale domains, the nonlinear minimizer beyond the coefficient used here, or an interacting Yang–Mills continuum limit.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b908671 70b30fe10e`. The exact deposited passages used for the coordinate and operator identifications are listed in the text.

[2] T. Bałaban, *Averaging operations for lattice gauge theories*, Commun. Math. Phys. 98 (1985), 17–51. Publisher record. This calculation uses the pinned deposited transcriptions; it does not claim a new full-text verification of the original paper.